

BANK CUSTOMER CHURN PREDICTION

A Hypothetical Case Study for JPMorganChase

Machine Learning & Business Analytics

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Independent hypothetical case study using a public synthetic Kaggle dataset.

Not affiliated with or endorsed by JPMorganChase.

01 Analytics Opportunities

Six analytics topics were considered for the same business goal: supporting informed, resource-constrained customer retention decisions.

01 Supervised machine learning

Use historical customer attributes and known churn labels to predict future churn risk; prioritize outreach.

02 Unsupervised machine learning

Cluster customers by balances, engagement and product ownership to design tailored retention programs.

03 Dimensionality reduction

Use PCA to compress correlated attributes and visualize customer patterns; test whether reduced features preserve signal.

04 Natural language processing

Analyze service complaints and messages to identify frequent problems and prioritize operational fixes.

05 Semantic and sentiment analysis

Identify dissatisfaction themes and sentiment in feedback to triage customer experience concerns.

06 Social media and digital analytics

Compare channel engagement and campaign conversion to decide where to invest outreach budgets.

Selected topic and rationale

Supervised machine learning is the best fit because the Kaggle training dataset includes a labeled binary outcome (Exited). Logistic Regression offers an interpretable starting model for ranking customer churn risk.

Decision maker

Hypothetical Customer Retention Manager at JPMorganChase. This is a business scenario only; the Kaggle data are synthetic and are not JPMorganChase customer records.

02 Machine Learning Overview

SUPERVISED MACHINE LEARNING

From labeled data to retention decisions

INPUT		MODEL		OUTPUT
Customer attributes	>	Logistic Regression	>	Churn probability

Origins & evolution

Statistical regression and probability models laid the groundwork for modern classification. Logistic regression became a standard method for binary outcomes; advances in computing and data availability expanded supervised learning across business applications.

Applications in banking and beyond

- Banking: flag customers at risk of leaving and rank outreach opportunities.
- Fraud: estimate likelihood that a transaction requires investigation.
- Healthcare: support risk screening using labeled clinical outcomes.

Business value

A ranked churn-risk list can help managers allocate a limited contact budget, monitor recall among churners, and compare targeted outreach against contacting everyone. Prediction alone does not establish that an intervention will prevent churn.

ILLUSTRATIVE CAMPAIGN ECONOMICS (NOT OBSERVED RESULTS)

1,000 contacts × $10 cost = $10,000

50 incremental saves × $400 contribution = $20,000

Illustrative net benefit: $10,000 | ROI: 100%

Connection to this project

Target: Exited (0 = stayed, 1 = exited). Compare Logistic Regression with a majority-class baseline; evaluate ROC-AUC, recall, precision and F1 on held-out labeled data.

METHOD EVOLUTION

1958	Later decades	Today
Cox: binary regression	Wider statistical adoption	Scalable ML workflows

Background: Cox (1958); scikit-learn documentation; Kaggle Playground Series S4E1.

03 Business Problem & Approach

Project title

Predicting Bank Customer Churn Using Machine Learning: A Hypothetical Case Study for JPMorganChase

Problem and business decision

Banks may lose future customer contribution when customers close accounts. A retention manager has limited staff time and campaign budget and must decide which customers to contact first. This project evaluates whether a transparent churn classifier can produce useful risk rankings for that decision.

Research question

How accurately can Logistic Regression predict bank customer churn using demographic, financial, and account-related characteristics, and how could these predictions support cost-effective customer retention decisions?

Stakeholder, use case and intended outcome

The hypothetical stakeholder is a Customer Retention Manager at JPMorganChase. The intended output is a validated churn-risk score and an example prioritized contact list. This is not an operational JPMorganChase system and uses no JPMorganChase data.

Proposed evaluation and success criteria

Compare Logistic Regression against a majority-class baseline on labeled data. Evaluate holdout ROC-AUC, churn Recall, Precision, F1 and a confusion matrix. Success means improved churn detection over the baseline, with useful precision-recall trade-offs for a limited outreach budget. Thresholds will be selected on validation data; no achieved performance is claimed yet.

Financial framing and limitations

A hypothetical campaign with 1,000 contacts costing $10 each and 50 incremental customer saves worth $400 each yields $10,000 illustrative net benefit. These inputs are assumptions, not observed campaign effects. A randomized retention test would be needed to estimate causal lift and actual ROI.

04 Dataset & Initial Evidence

Data provenance and scope

Source: Kaggle Playground Series S4E1, Binary Classification with a Bank Churn Dataset (synthetic competition data). Labeled train.csv: 165,034 rows and 14 columns, including Exited; unlabeled test.csv: 110,023 rows and 13 columns. Each row is a synthetic customer-like observation, not a verified unique real customer. CustomerId repeats, so identifier handling and possible leakage require care.

Initial quality checks

EDA found no missing values or exact duplicate rows. Repeated CustomerId values require care. Public synthetic Kaggle data are not JPMorganChase customer records; competition data-use terms should be checked before redistribution.

INITIAL INSIGHT

Candidate predictors: Age, CreditScore, Balance, Geography, Gender, Tenure, NumOfProducts, IsActiveMember, EstimatedSalary. Target: Exited (0/1).

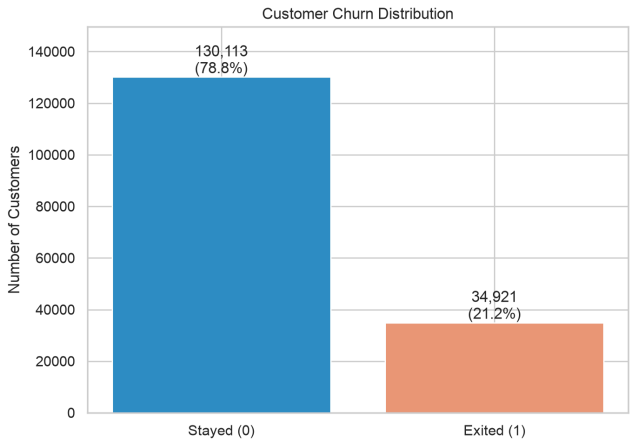


Figure 1. Overall churn

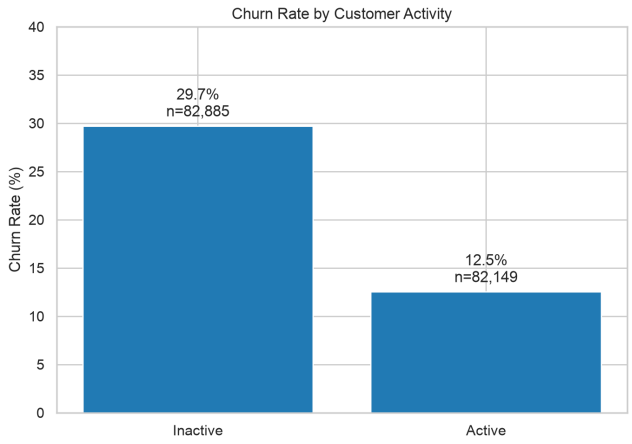


Figure 2. Activity

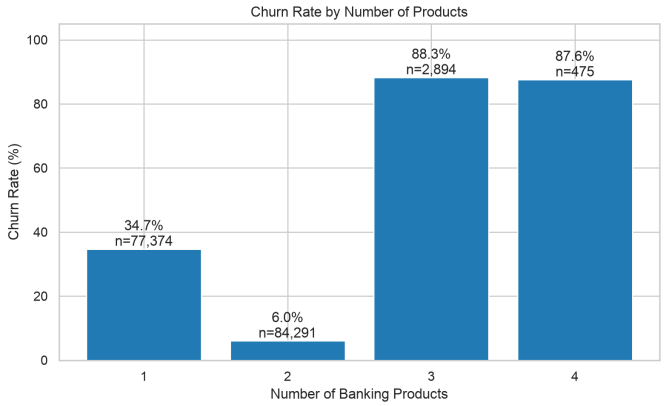


Figure 3. Product count

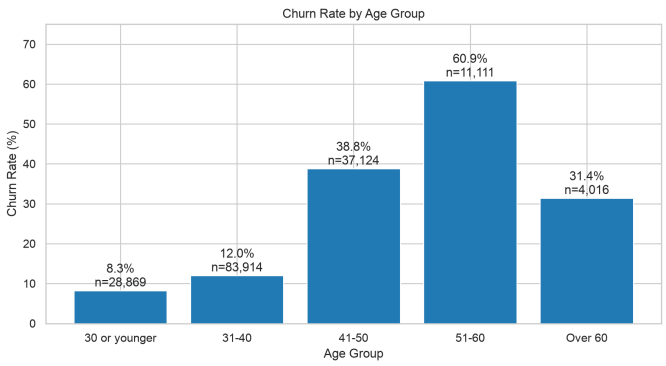


Figure 4. Age groups

Figures generated from the labeled Kaggle training data; descriptive comparisons only.

05 Findings & Modeling Roadmap

What the four figures show

• Overall churn: 34,921 exited (21.2%) versus 130,113 stayed (78.8%); accuracy alone may be misleading.

• Activity: inactive observations show 29.7% churn (n=82,885) versus 12.5% for active (n=82,149).

• Products: churn is 34.7% with one product and 6.0% with two; 3 and 4 products show 88.3% and 87.6%, but these groups are smaller (n=2,894 and 475).

• Age: churn peaks at 60.9% among ages 51–60 (n=11,111), versus 8.3% at age 30 or younger (n=28,869).

Modeling plan and baseline

Remove identifiers and other inappropriate predictors; encode categorical variables using training-only preprocessing; stratify labeled data into development and untouched holdout subsets. Check whether repeated CustomerId could cause leakage across splits and document a conservative split strategy. Fit a majority-class predictor as baseline, then Logistic Regression. Select any threshold on validation data only; report final metrics on the holdout.

Feasibility and risks

The labeled sample is large enough for an initial classification experiment and initial charts have been produced. Key risks include synthetic-data generalizability, class imbalance, correlated observations, subgroup instability, and interpreting predictive associations as causal effects. No trained-model results are asserted at this stage.

Next steps

Finalize leakage-safe split and feature preprocessing; train and validate the baseline and Logistic Regression; generate ROC curve and confusion matrix; document reproducible run instructions; discuss targeting trade-offs and the need for experimental validation of campaign impact.

References

Kaggle (2024). Playground Series S4E1: Binary Classification with a Bank Churn Dataset.
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Cox, D. R. (1958). The regression analysis of binary sequences. Journal of the Royal Statistical Society: Series B, 20(2), 215–242.
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https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

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https://scikit-learn.org/stable/modules/model_evaluation.html

BASELINE FEASIBILITY CHECK | DERIVED FROM CLASS COUNTS

Always predicting "Stayed" gives 130,113 / 165,034 = 78.8% in-sample accuracy.
Churn recall = 0%. This is arithmetic, not a trained or holdout-tested model result.